Asteroid Flybys from the SEL2 2nd Lagrange Point

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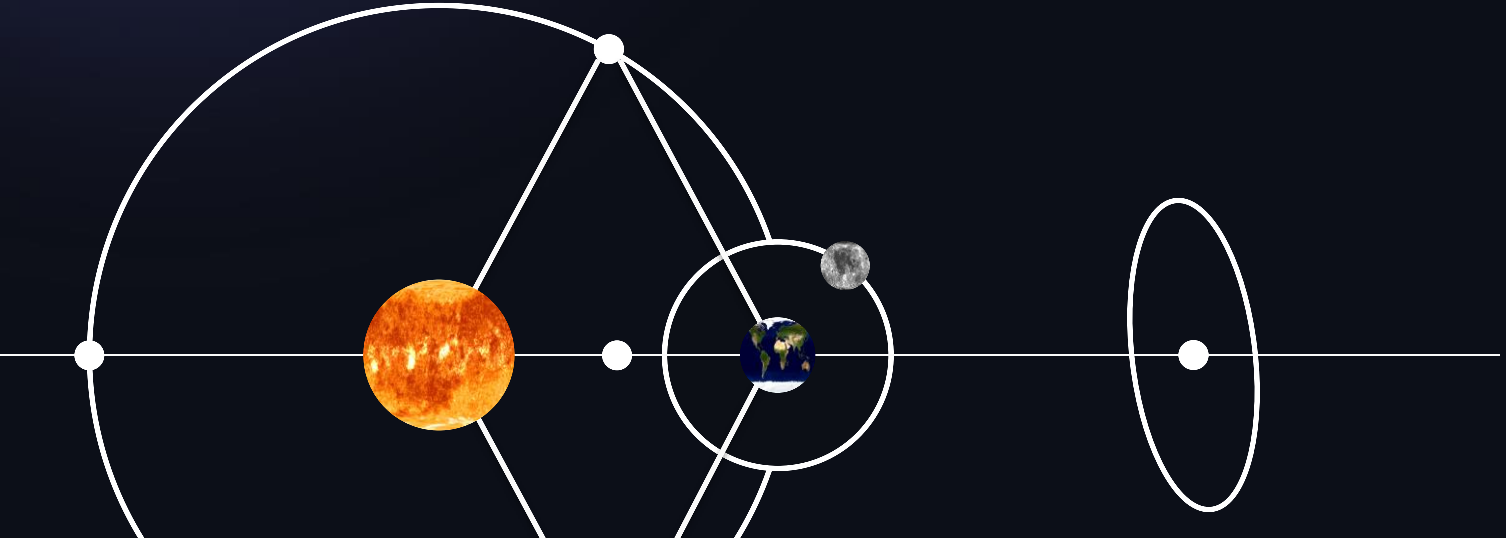

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Project Background

The Sun-Earth 2nd Lagrange Point (SEL2)

- The SEL2 is popular for its stable thermal environment, making it great for observatories and probes alike:
  - Think Gaia, James Webb, PLATO, and more…

- For Primary missions to SEL2, the delta-v may well be used well within the margins during the launch dispersion stages.

- If there is fuel left – what next?
  - Is it wasted alongside decommission?
  - Or can we use this for secondary science?

Objectives

Goal & Constraints

~ With the minimal fuel remaining from a Primary mission, demonstrate it is possible to complete an asteroid flyby from the SEL2 point. ~

Mission Requirements

[ESOC, 2022]

- **SEL2 Departure window: 2025-2027**
- **Delta-v $\leq$ $150 ms^{-1}$**
- **Relative Arrival Velocity $\leq 5 kms^{-1}$**
- **TOF $\lesssim$ 9 months**

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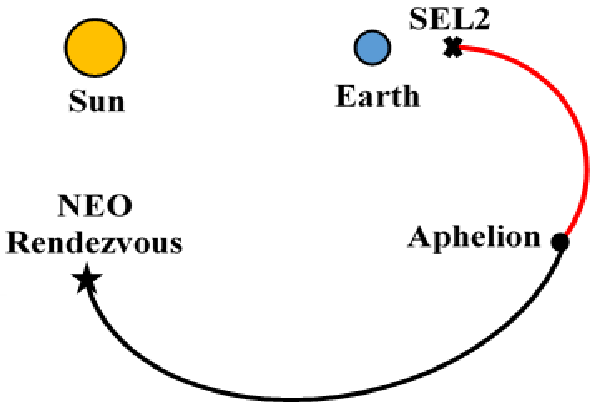
Methodology

A Top-Level Look

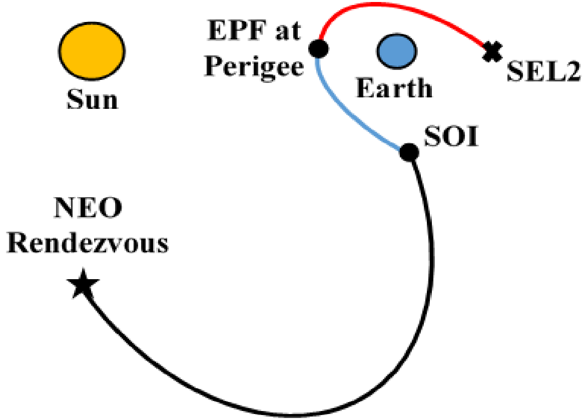
Mission Scenario 1

1. Locate the 2nd Lagrange Point
2. Manifold transfer to the Aphelion
3. Prune asteroids with MOID
4. Build Pork Chop Plots using Lambert
5. Optimise for mission constraints



Mission Scenario 2

1. Locate the 2nd Lagrange Point
2. Manifold transfer to the Perigee for EPF
3. Prune asteroids with MOID
4. Build Pork Chop Plots using Lambert
5. Optimise for mission constraints



Circular Restricted 3 Body Problem (1)

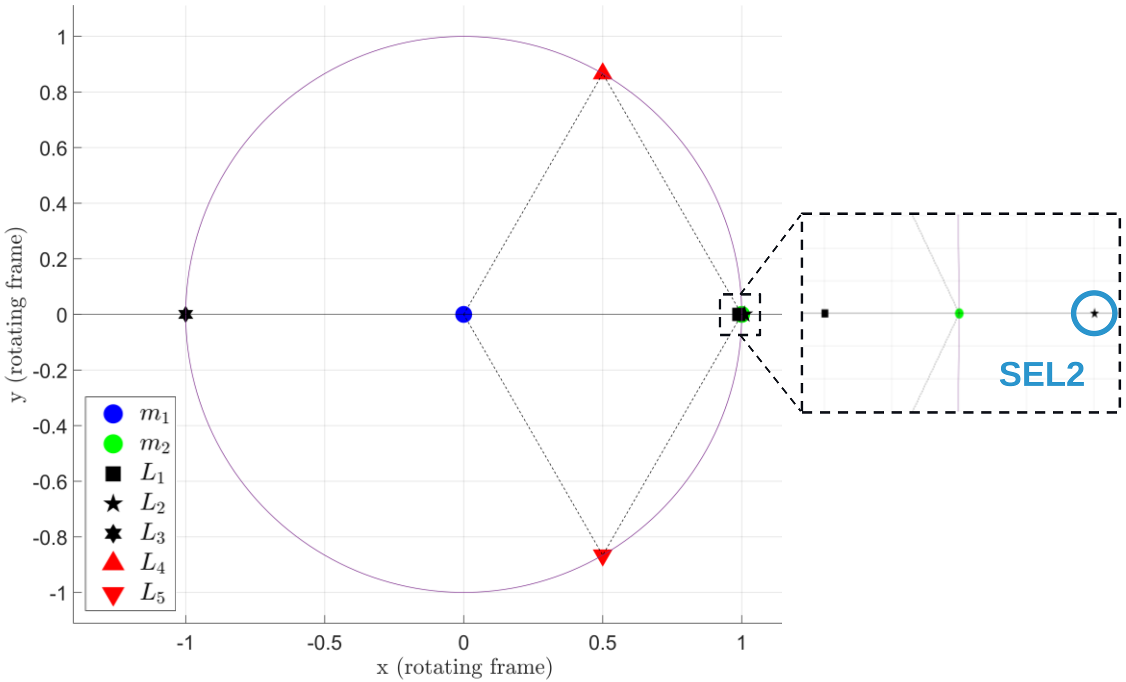
Locating SEL2

Assumptions / Set-up:

- Primaries: Sun and Earth (circular, Keplerian)
- Spacecraft mass negligible
- 2 Ref. frames: Inertial and Synodic
- Normalised Units (NU)
    - $1\ LU = a\ (AU)$
    - $1\ TU = 1/n$

- SEL2 located as point vector using Newton's method

- Jacobi Integral used as validation



Circular Restricted 3 Body Problem (2)

Escaping SEL2

Perturbation:

$$IC_{1,2} = X^* \pm \varepsilon V_{unstable}$$

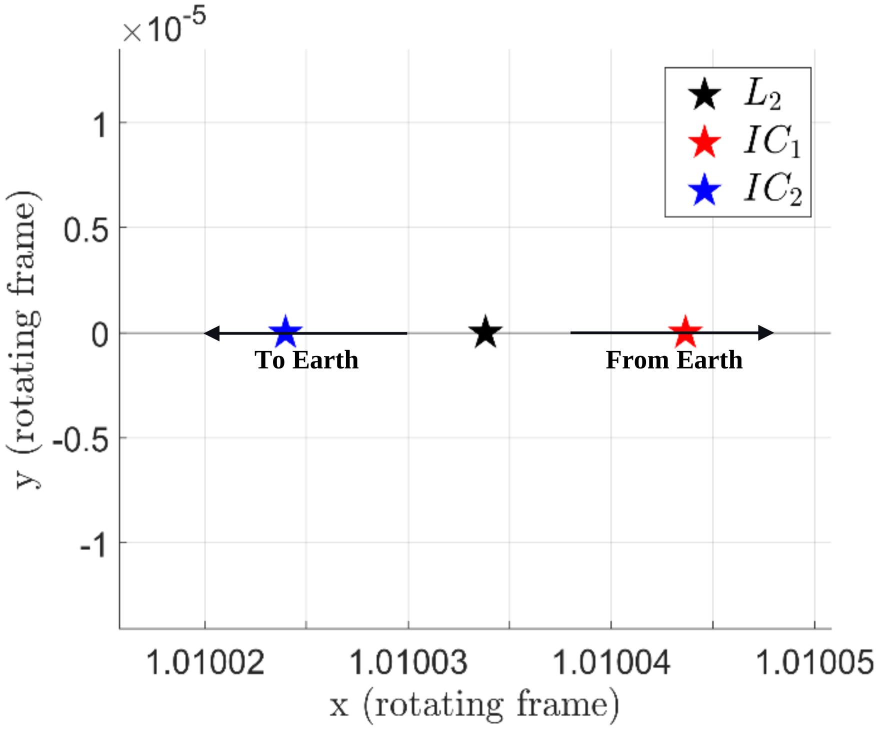
Manifold Transfer:

- **Propagate from SEL2 via CR3BP**
  - **MS 1, to exterior realm**
  - **MS 2, to interior realm**
- **Terminate at orbit entry event:**

$$\vec{r}_{wrt} \cdot v = 0$$

Leg 1:

- **MS 1: 343 days (~11.4 months)**
- **MS 2: 152 days (~5.0 months)**



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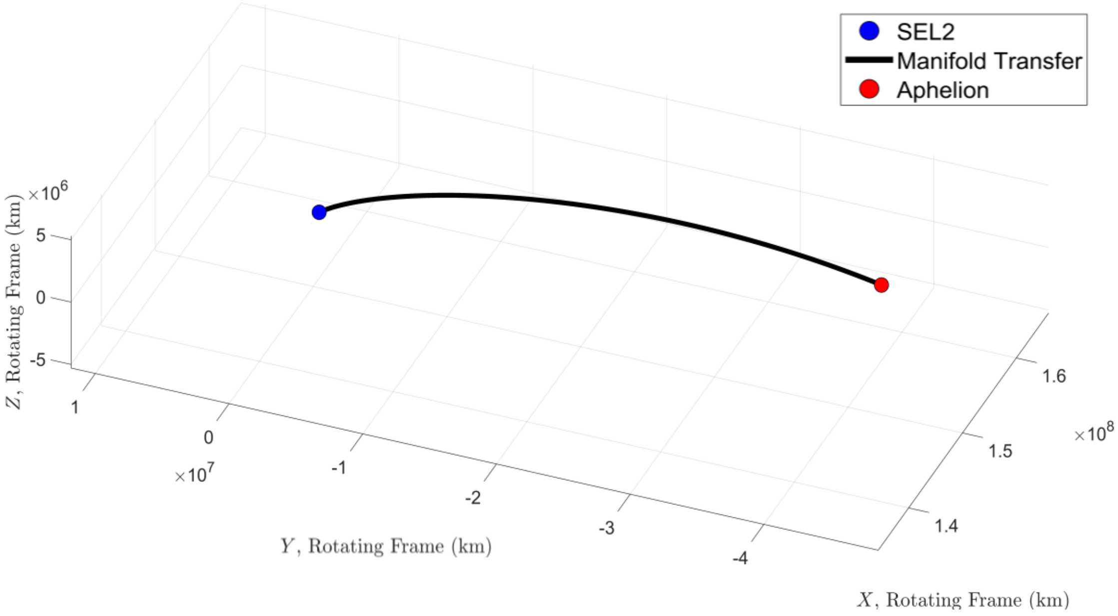

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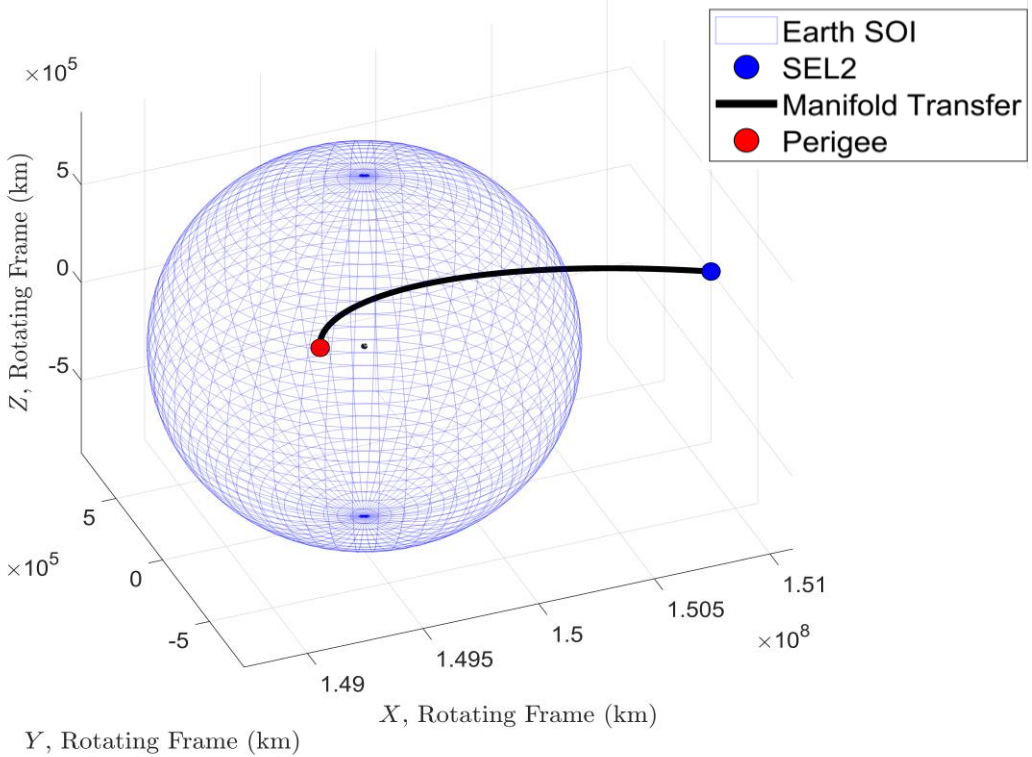
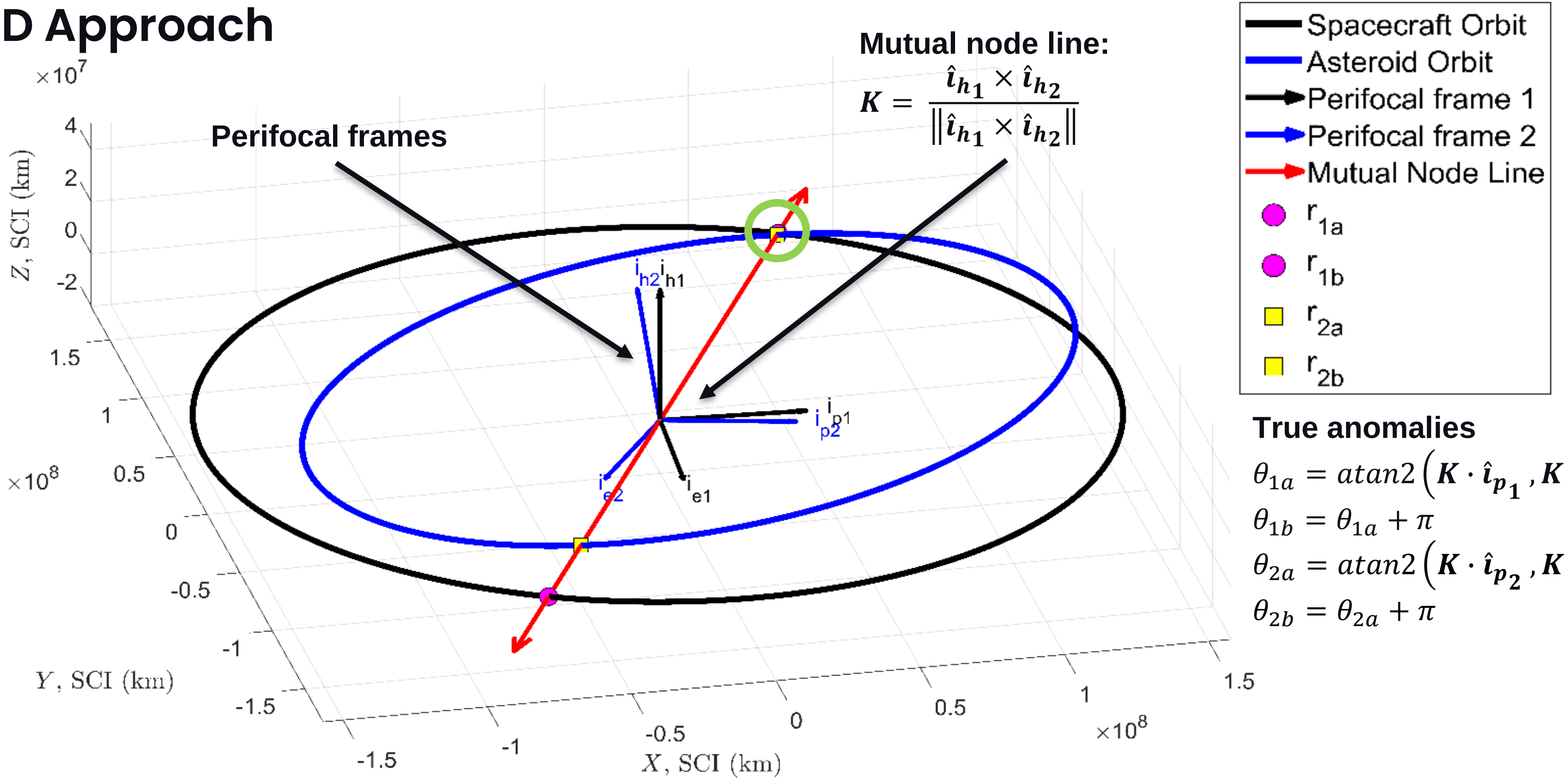


Asteroid Searches (1)

MOID Approach

Mutual node line:

$$\boldsymbol{K} = \frac{\hat{\boldsymbol{i}}_{h_1} \times \hat{\boldsymbol{i}}_{h_2}}{\left\| \hat{\boldsymbol{i}}_{h_1} \times \hat{\boldsymbol{i}}_{h_2} \right\|}$$



True anomalies

$$\theta_{1a} = atan2\left(\boldsymbol{K} \cdot \hat{\boldsymbol{i}}_{p_1}, \boldsymbol{K} \cdot \hat{\boldsymbol{i}}_{e_1}\right)$$
$$\theta_{1b} = \theta_{1a} + \pi$$
$$\theta_{2a} = atan2\left(\boldsymbol{K} \cdot \hat{\boldsymbol{i}}_{p_2}, \boldsymbol{K} \cdot \hat{\boldsymbol{i}}_{e_2}\right)$$
$$\theta_{2b} = \theta_{2a} + \pi$$

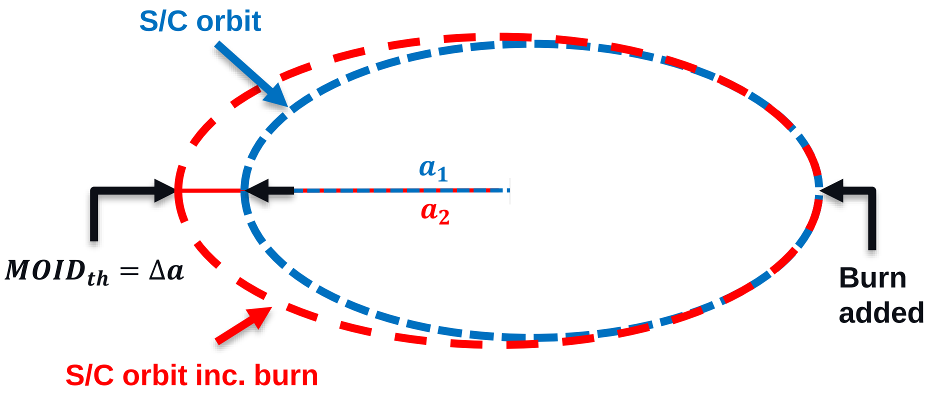
Asteroid Searches (2)

Asteroid Search from Aphelion – MS 1

Approach:

- Rotate from Synodic to Ecliptic J2000 frame
- Filter out NEOs: $i < 10°$ && H $\lesssim$ 25

- If Distance, $MOID < MOID_{th}$ (0.0129 AU)
  - Build Pork chop Plot
- If minimum delta-v trajectory has:
  - Departure Velocity, $v_{dep} \lesssim 160ms^{-1}$
  - Arrival Velocity, $v_{rel} \lesssim 6kms^{-1}$
- Store NEO and trajectory data



Found:

- 71 NEOs returned
- 31 meeting velocity constraints

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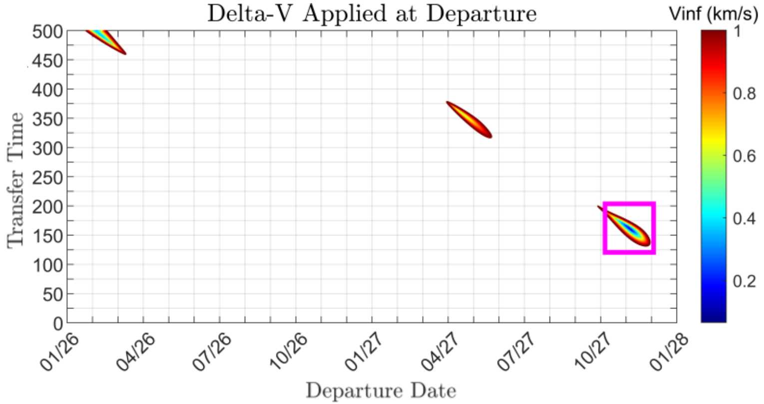
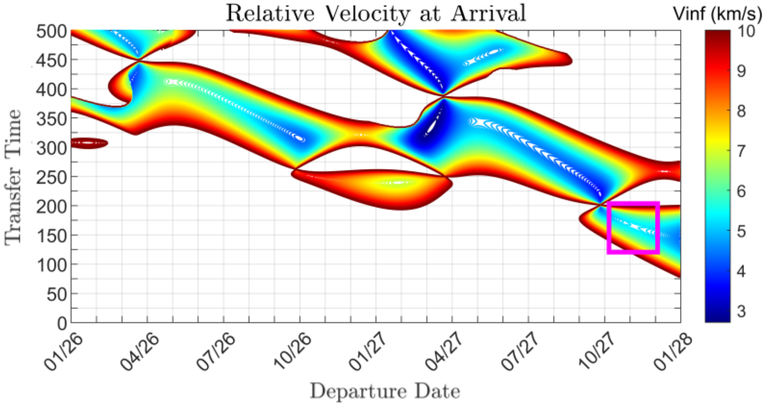
Found:

- 71 NEOs returned (10 optimised)
- 31 meeting velocity constraints





Asteroid Searches (3)

Asteroid Search from Perigee – MS 2

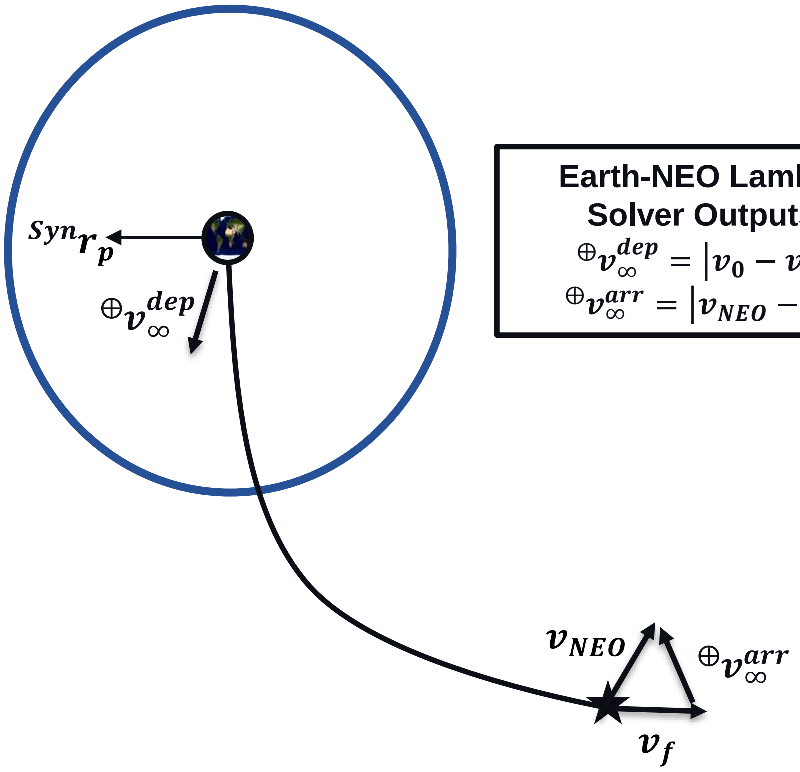
Approach:

- Rotate from Synodic to Ecliptic J2000 frame
- Filter out NEOs: $i < 10°$ && H $\lesssim$ 25

- If Distance, $MOID < MOID_{th}$ (0.12701 AU)
  - Build Pork chop Plot
    - Lambert solver – Earth to NEO
    - Escape trajectory from Perigee to SOI
- If minimum delta-v trajectory has:
  - Departure Velocity, $v_{dep} \lesssim 300ms^{-1}$
  - Arrival Velocity, $v_{rel} \lesssim 6kms^{-1}$
- Store NEO and trajectory data

From MS 2 EPF:

- 20 NEOs returned



Earth-NEO Lambert Solver Outputs:

$^{\oplus}v_{\infty}^{dep} = |v_0 - v_{\oplus}|$

$^{\oplus}v_{\infty}^{arr} = |v_{NEO} - v_f|$

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Asteroid Searches (3)

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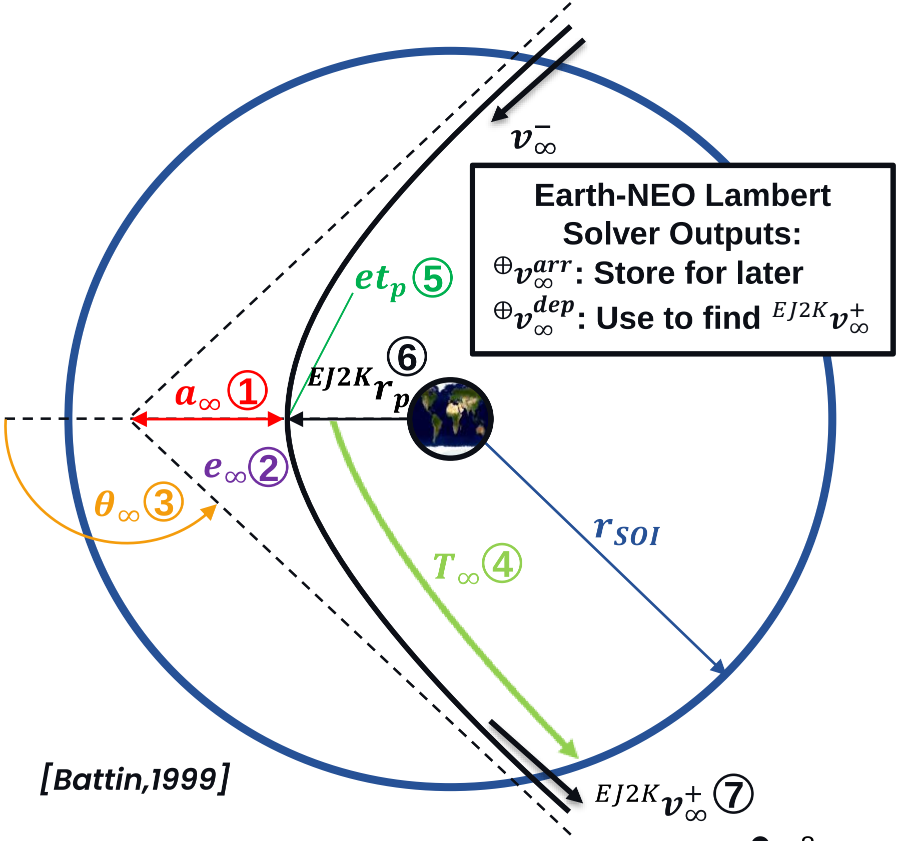


[Battin,1999]

Trajectory Optimisation (1)

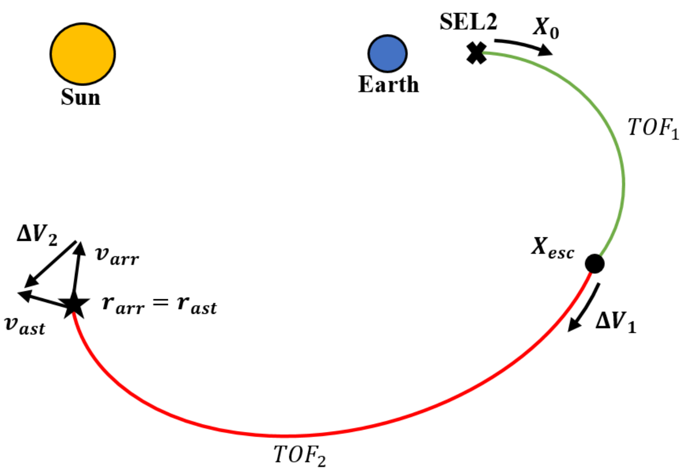
Set-up for Fmincon

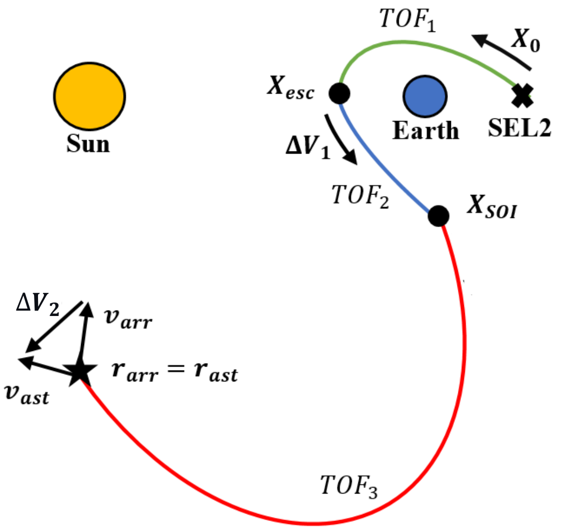
Cost Functions:

- Departure delta-v
- Total TOF
- Relative arrival delta-v

Constraints

- Same start point (SEL2)
- $r_{s/c} - r_{ast} = 0$ at arrival
- $v_{s/c} + \Delta V_2 - v_{ast}$ at arrival
- At/Outside SOI (MS 2)
- Aphelion/Perigee entry (removable)
- $\|\Delta V_1\| \leq 150ms^{-1}$
- $\|\Delta V_2\| \leq 5kms^{-1}$





Trajectory Optimisation (2)

MS 1 Results

- **Aphelion entry lifted**
- **All NEOs below delta-v**
- **8/10 below relative arrival velocity**

Name	TOF_{tot} $(Days)$	TOF_{tot} $(Mo.)$	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})	Name	TOF_{tot} $(Days)$	TOF_{tot} $(Mo.)$	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})
2004 FM17	502.13 [-3.18]*	16.74 [-0.11]	0.1500 [+0.0896]	5.3282 [+0.0493]	2013 XV8	376.22 [+0.91]	12.54 [+0.01]	0.1500 [+0.0116]	3.3616 [-0.1340]
2004 DK1	558.78 [-4.53]	18.63 [-0.15]	0.1500 [+0.0993]	5.4206 [-0.1025]	2005 TF45	382.25 [-2.06]	12.74 [-0.07]	0.1500 [+0.0457]	3.8592 [+0.1248]
2014 OA2	606.60 [+0.29]	20.22 [+0.01]	0.1500 [+0.1122]	4.9677 [-0.0515]	2014 OV3	422.63 [-2.68]	14.09 [-0.09]	0.1500 [+0.0228]	4.5634 [+0.1229]
2015 DP155	639.28 [-4.03]	21.31 [-0.13]	0.1499 [+0.1384]	4.2687 [-0.0973]	2024 CN5	429.48 [-4.83]	14.32 [-0.16]	0.1500 [+0.1263]	4.247 [+0.1052]
2009 CO5	797.59 [+0.28]	26.59 [+0.01]	0.1499 [+0.0289]	4.9666 [-0.0175]	2008 MQ1	451.25 [+3.94]	15.04 [+0.13]	0.1499 [+0.0105]	4.3766 [-0.0545]

*Increase [+] or decrease [-] from original trajectories. Un-highlighted results were optimised for TOF. Highlighted results were optimised for relative velocity at arrival.

Trajectory Optimisation (3)

MS 2 Results

- Perigee entry lifted
- Half of initial guesses converged
- 6/10 above 500ms^{-1} (removed)
- 1 with delta-v below $150 ms^{-1}$
- 2 below relative arrival velocity
- 2 also close to TOF constraint

Name	TOF_{tot} $(Days)$	TOF_{tot} $(Mo.)$	v_{∞}^{dep} (kms^{-1})	v_{∞}^{arr} (kms^{-1})
2016 HO3	321.71 [+31.39]*	10.72 [+1.05]	0.2567 [+0.0126]	5.1457 [-0.0568]
2022 QM6	295.51 [+30.77]	9.85 [+1.03]	0.4604 [+0.1855]	5.7791 [+0.3924]
2021 RS	413.1047 [+16.52]	13.77 [+0.55]	0.1300 [-0.1311]	2.441 [+0.1346]
2015 RF36	457.57 [+15.72]	15.25 [+0.52]	0.2083 [-0.0694]	4.2703 [+0.4227]

*Increase [+] or decrease [-] from original trajectories

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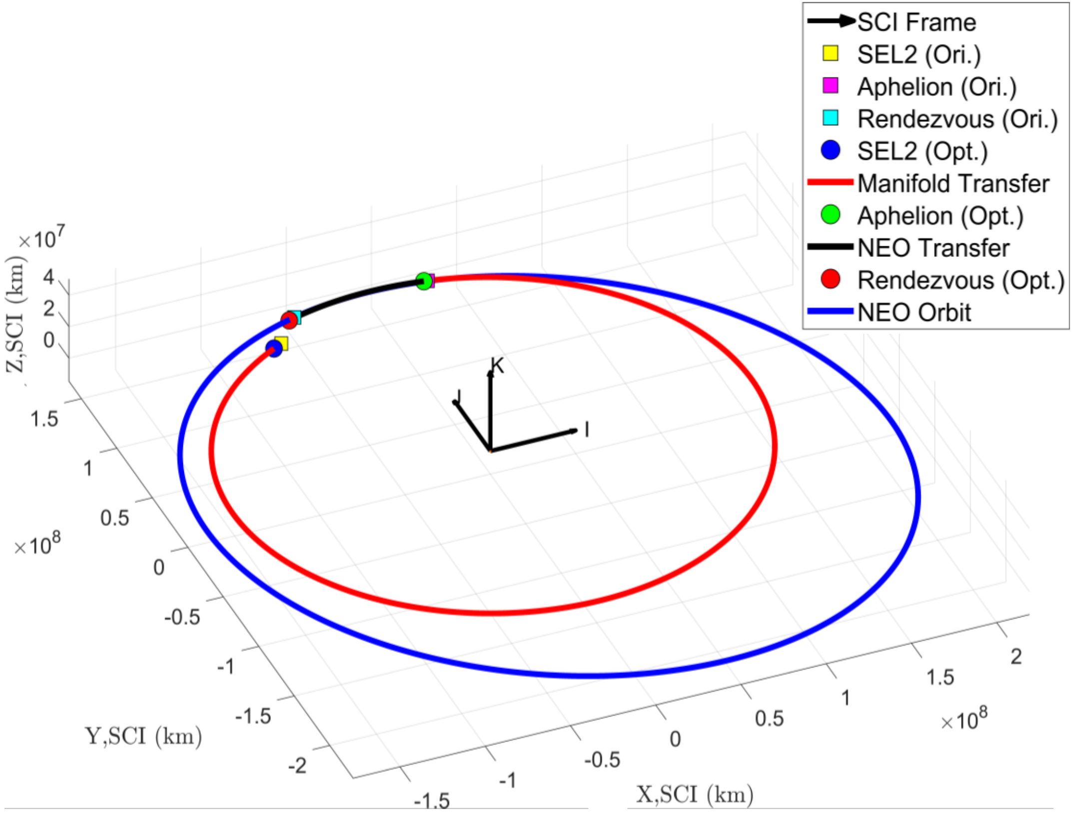
Findings

Currently:

MS	Name	Abs. Mag.	TOF_{tot} (Mo.)	v_∞^{dep} (kms^{-1})	v_∞^{arr} (kms^{-1})
1	2004 FM17	19.36	16.74	**0.1500**	5.3282
1	2004 DK1	21.30	18.63	**0.1500**	5.4206
1	2014 OA2	21.48	20.22	**0.1500**	**4.9677**
1	2015 DP155	21.62	21.31	**0.1499**	**4.2687**
1	2009 CO5	21.85	26.59	**0.1499**	**4.9666**
1	2013 XV8	21.89	12.54	**0.1500**	**3.3616**
1	2005 TF45	23.10	12.74	**0.1500**	**3.8592**
1	2014 OV3	23.20	14.09	**0.1500**	**4.5634**
2	2015 RF36	23.40	15.25	0.2083	**4.2703**
2	2022 QM6	23.74	**9.85***	0.2567	5.7791
1	2008 MQ1	24.20	15.04	**0.1499**	**4.3766**
2	2016 HO3	24.33	**10.72***	0.2567	5.1457
2	2021 RS	24.94	13.77	**0.1300**	**2.441**
1	2024 CN5	24.98	14.32	**0.1500**	**4.2470**



Limitations of Study

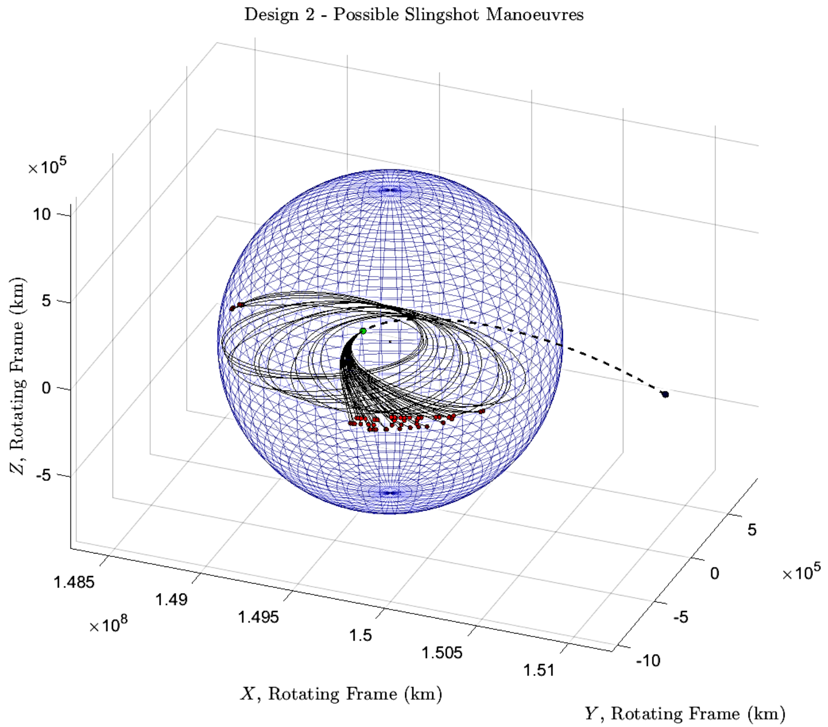
Observations

MS 1:

- **SEL2 modelled as point vector**
- **TOF constraint out of reach with current 1st leg**

MS 2:

- **SEL2 modelled as point vector**
- **Hyperbolic EPF initial guess appears to be infeasible**
  - **Result of a computational trade-off**
- **Optimisation of MS 2 is highly sensitive**



What’s Next?

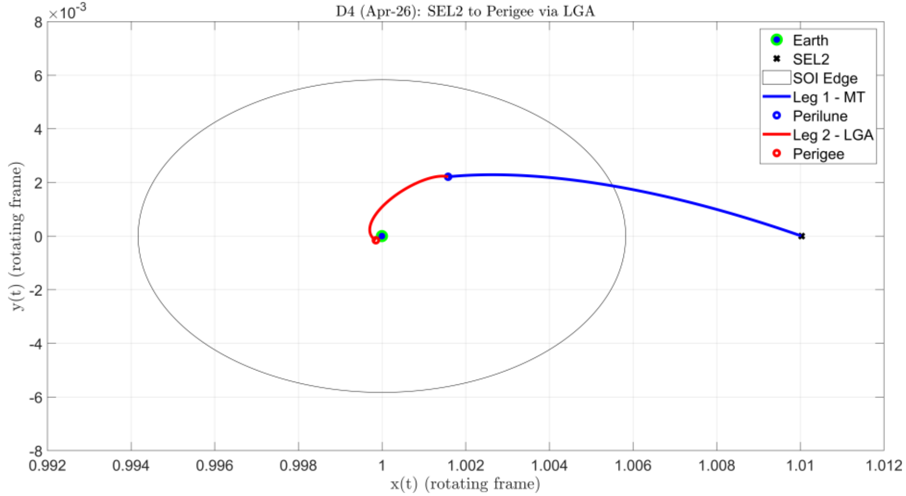
Plenty!

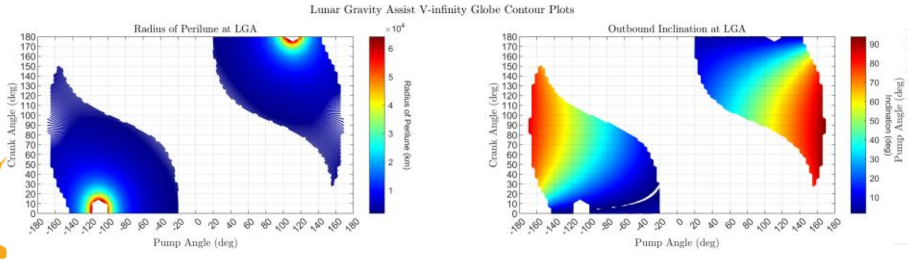
On current designs:

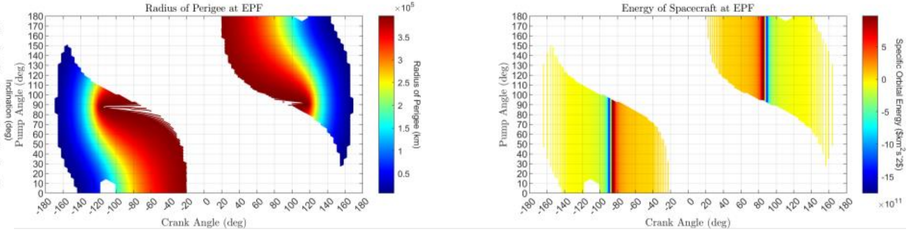
- **MS 1 asteroid search could be relaxed**
- **Improve MS 2 initial guess**

Looking forward:

- **What about an LGA on its own?**
- **Do additional flybys help?**
- **Can we lower the Perigee for EPF?**
- **Can we do multiple rendezvous?**







Conclusions

Findings:

- 91 preliminary candidate NEOs found across 2 mission scenarios
- 11/30 optimised reachable within the delta-v limit
- 9/11 of these also within the relative arrival velocity
- 2 in the region of a 9 month TOF

Scientific Contributions

- Purely geometric MOID approach demonstrated
- Successfully shown that leftover fuel is recyclable from SEL2

Not only can asteroid flybys be achieved for a realistic leftover fuel, but they can also be completed for significantly less if desired.

Thank You
For Your Attention

Any Questions?

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For more information, please see the full paper:

Asteroid Flybys From The L2 Sun-Earth Lagrange Point
IAC-24,C1,6,10,x85519